Residual Dynamics Learning Under Partial Governing Equations in Physics-Informed Neural Networks

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A. Motivation and Background

Physics-Informed Neural Networks (PINNs) incorporate known governing equations into training by penalising violations of those equations at collocation points. The approach works well when the governing equations are known correctly. The difficulty is that in many systems of genuine scientific interest, they are not: equations are simplified, forcing terms are dropped for tractability, or certain interactions are simply unknown. A turbulence model omits sub-grid-scale stresses. A biological signalling model omits unmeasured reaction pathways. In each case, something is missing from the physics, and the effect of that omission on training is not well understood.

If the missing terms are small, the PINN may still learn a reasonable solution, with the data loss compensating for the physical gap. If the missing terms are large, enforcing the wrong equations may actively harm learning — pushing the model toward solutions that satisfy the incomplete law but deviate from the true dynamics. I am genuinely uncertain which regime applies in practice, and under what conditions the transition occurs. That uncertainty is the starting point for this proposal.

B. Literature Understanding

Raissi, Perdikaris, and Karniadakis [1] introduced the standard PINN formulation and showed it can solve both forward and inverse PDE problems with limited data by treating the PDE residual as a regularisation term. Their benchmark problems all assume the governing equations are fully known. Subsequent work has identified several ways PINNs can fail even under that assumption: networks preferentially fit low-frequency solution components first [2], and competing gradient signals between the data and physics loss terms can prevent convergence, particularly in problems with sharp gradients [3]. These failure modes are relevant background because they suggest that PINNs are already sensitive to the structure of the physics constraint — which motivates asking what happens when that structure is degraded.

On the side of hybrid physics–data models, Levine and Stuart [4] study a setting where a known dynamical model is augmented with a learnable correction, and show that under certain conditions the correction can absorb model error without corrupting the physics-based component. Their analysis is for ODE systems and does not examine how behaviour changes as the known model becomes progressively less accurate. Chen et al. [5] showed that residual networks can be interpreted as continuous dynamical systems, providing some grounding for the idea that a neural correction term can stand in for missing dynamics. What neither contribution addresses is the intermediate PDE regime — where some physics is available but some is missing — or how training stability varies across that spectrum.

C. Proposed Direction

Central question: Does a residual neural correction term stabilise or destabilise PINN training when the governing equation is incomplete, and does the answer depend on how much physics is missing?

This question is narrower than it might sound. I am not asking whether residual correction improves prediction accuracy in general — that is already known to be possible. I am asking specifically about the interaction between the correction term and the incomplete physics constraint during training. My intuition is that a small degree of incompleteness is relatively benign: the residual network fills in the missing dynamics and the physics constraint still provides useful structural guidance. But beyond some threshold, the constraint may begin to mislead rather than guide, and the residual network has to simultaneously compensate for the bad constraint and learn the true dynamics. Whether this produces a clean transition in training behaviour, or a gradual degradation, is what I want to find out.

The novel aspect of this direction is treating physical completeness as a continuous variable rather than a binary one. This is closer to the reality of applied scientific modelling, where equations are almost always approximate rather than simply right or wrong.

D. Methodological Intuition

The test system will be a family of forced viscous Burgers equations:

$$\partial u/\partial t + u\,\partial u/\partial x = \nu\,\partial^2 u/\partial x^2 + f(x, t)$$

where f(x, t) is a known forcing term present in the true system but withheld from the PINN's physics constraint. Three forcing types will be used — sinusoidal, polynomial, and a spatially localised pulse — to check whether findings are robust across different missing-term structures. Burgers is a natural choice: it is nonlinear, admits shocks, and is cheap to simulate accurately, so ground truth is easy to generate.

Physical incompleteness is controlled by a scalar $\alpha \in [0, 1]$, where $\alpha = 1$ means the full equation is enforced and $\alpha = 0$ means only the homogeneous Burgers term is enforced. The PINN physics loss is:

$$L_{phys} = \| \partial u_\theta/\partial t + u_\theta\,\partial u_\theta/\partial x - \nu\,\partial^2 u_\theta/\partial x^2 - \alpha \cdot f - r_\varphi \|^2$$

where u_θ is the solution network and $r_\varphi(x, t)$ is a small auxiliary network representing the residual correction; θ and φ are their respective trainable parameters. Four models are compared at each α: (i) full-physics PINN ($\alpha = 1$, no r_φ), (ii) partial-physics PINN without correction, (iii) partial-physics PINN with correction, and (iv) a purely data-driven network. Ground truth is generated by a Fourier pseudospectral solver. An α sweep of ten values across three forcing types and four model variants gives 120 training runs. Burgers in 1+1 dimensions with a few thousand collocation points converges in minutes on a single GPU, making this feasible within an internship.

Primary diagnostics: solution error on held-out trajectories, training loss curves as a function of α, and — where possible — comparison of learned r_φ against the true omitted forcing, to check whether the correction is learning something physically meaningful or absorbing noise.

E. Expected Challenges

The hardest problem is separating genuine residual correction from overfitting. The auxiliary network r_φ is unconstrained, so it could memorise the training data rather than learning the missing dynamics. Evaluating on held-out initial and boundary conditions and comparing r_φ to the known omitted forcing are the planned mitigations, though I am not sure they will be fully conclusive — there may be cases where r_φ fits training data well and resembles the true forcing without this being evidence of generalisation.

A second difficulty is that incomplete physics may worsen the loss landscape. PINNs already suffer from gradient imbalance between physics and data terms when the physics is correct [3]; an incorrect physics constraint may amplify this. Adaptive loss weighting [7] or gradient normalisation [3] will likely be necessary, and choosing the right approach for each value of α may itself require tuning I do not yet know how to automate.

The results will also be specific to Burgers and may not carry over to systems with richer structure. I regard this as an acceptable scope limitation: a clean result on a canonical system is more useful than a noisy result on a complex one. Finally, I should be honest that my current understanding of PINN training dynamics comes mainly from reading rather than implementation, and I expect a learning curve in the early weeks.

F. Potential Applications

The most direct application is turbulence closure. In large-eddy simulation, the Navier–Stokes equations are solved on a coarse grid and the effect of unresolved small-scale eddies enters through a closure term that is not derived from first principles. This is exactly the structure of the proposed setup: a known PDE with a missing term that a learnable correction is meant to supply. If this investigation establishes when residual correction is stable and when it is not, that has direct implications for how neural closure models should be trained alongside the resolved equations.

A second application is in systems biology. Models of intracellular signalling — phosphorylation cascades, gene regulatory networks — are routinely incomplete because not all reaction partners are known or measurable. A PINN-based approach that works with a partial mechanistic model and a learnable correction for unknown interactions would be useful in this setting, provided the training behaviour is well enough understood to trust what the correction has learned.

G. References

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